Magnum Bones Translation Atlas

Protocol for tracing orthopaedic research into patient care and benefit

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Systematic evidence map with a nested probability sampled report cohort and systematic reviews of linked clinical evidence

Purpose

This study will determine which identifiable contributions from musculoskeletal basic science, biomechanics, materials research and computational modelling are subsequently used, what changes during translation, and what evidence supports their use in patients. Its central clinical question is whether documented patient exposure is accompanied by credible evidence of benefit and adequate assessment of harm. Useful negative experiments and documented discontinuation will be retained as consequential research outcomes.

The protocol specifies a broad bibliographic map, intensive tracing of 16,000 randomly selected records, and complete assessment within prespecified searches of the clinical destinations linked to eligible sampled reports. The number selected is a planned workload, not a result or a guarantee of clinical events. No new cohort has been drawn under this protocol.

Design in brief

Component	Population and purpose
Bibliographic map	Accessible, deduplicated reports published in 2005–2025; metadata from 2000 onward supports baseline field measures. Describes the searchable literature, not every eligible invention.
Mature report cohort	12,000 records first published in 2005–2015, randomly selected for eligibility assessment and five and ten year documentary follow up.
Recent report cohort	4,000 records first published in 2016–2020, randomly selected for five year follow up. Kept separate from mature ten year estimates.
Clinical destination review	For each verified destination of an eligible sampled report, identify all eligible clinical studies within the locked search and assess implementation, benefit and harms.

Primary outcome: contribution-linked patient management documented within ten years in the mature cohort. Performed research use, research-stage clinical implementation, routine-care documentation, comparative benefit, harms, authorization and informative negative findings remain separate outcomes. The primary denominator is eligible research reports; it is not unique inventions.

This is a dated extension after results from an earlier cohort were known. The extension will be publicly timestamped before its new sample is drawn. Registration has not yet occurred. The existing manuscript and dataset remain intact. Journal acceptance and the size or direction of findings cannot be specified by a protocol.

1 Rationale and contribution

Orthopaedic innovation reaches patients through several routes. A laboratory mechanism can guide a critical experiment; a computational model can change implant design; a procedure can enter practice without following a drug-development pathway. Uptake, commercial availability, regulatory permission and patient benefit are therefore different quantities. Treating them as one ladder conceals both productive research and premature adoption.

The novelty claim will be specific. An umbrella review already assessed 367 animal-tested therapies across 122 reviews.[1] A 2025 orthopaedic review examined 23 reports with biomechanical and clinical comparisons, and a 2026 review assessed clinical readiness of musculoskeletal AI.[2,3] These studies make a generic claim to discover the bench-to-clinic gap untenable. A BMJ review of five implant innovations also provides a direct precedent for examining evidence accompanying clinical adoption.[4]

The proposed advance is to start from a defined research denominator, preserve untraced beginnings, identify the exact contribution used downstream, and compare evidence available at documented clinical use with evidence accumulated later. The study will measure changes in the contribution, distinguish negative testing from inactivity, and expose how source availability changes inference. It will not infer efficacy from uptake or compare report-based percentages directly with therapy-based approval percentages.

What the existing work contributes

The saved manuscript dated 21 October 2026 contains 3,000 selected records: 471 eligible primary-frame reports and 122 eligible extension reports, with 202 records of uncertain eligibility. It documents 99 ten-year research links and three patient-management links in the primary frame. Its extensive unresolved evidence prevents a credible single clinical translation rate. These are existing development findings, not results of the new protocol. The source documents are the saved manuscript version 8 and supplement version 6.

The development cohort will supply vocabulary, failure cases, source-access restrictions and test cases for the extraction workflow. Its deterministic selection and nominal weights will not be relabelled as a probability sample. Previous results may inform this protocol; that influence is disclosed. Old analyses will be reported separately.

What changes in the extension

The extension replaces deterministic ranking with a genuine random draw, adds predeclared follow-up and source-completeness rules, makes patient management the single primary endpoint, requires direct ancestry for primary positive links, and adds structured clinical benefit and harm reviews. It expands the publication period and preserves a separate recent cohort. Reporting will use PRISMA-P, PRISMA 2020 and PRISMA-S for the relevant review components, with STROBE items for the sampled cohort.[5–7] Reporting checklists do not certify the validity of an automated assessment.

No “first”, “largest” or “most comprehensive” claim will appear without a reproducible comparison supporting that exact claim. The manuscript’s conclusion will follow the locked results, including a result dominated by uncertainty.

2 Questions and estimands

Priority	Question	Quantity
Primary	How often is a defined contribution subsequently used in patient management?	Ten year weighted report-level documented-use proportion and compatible uncertainty bounds in the mature cohort.
Secondary	How often does research inform performed research?	Five and ten year contribution-linked research use, including extension, replication and critical testing.
Secondary	What accompanies patient exposure?	Clinical study design, independent replication, benefit, harms and surveillance available before documented exposure, and by the horizon.
Secondary	What crosses the boundary?	Exact use versus documented adaptation; species, material, dose, geometry, algorithm, indication and procedural changes.
Secondary	Which signals differ from implementation?	Paired classifications of clinical citations, patents, commercial statements, authorization and actual patient management.
Secondary	Where does uncertainty arise?	Eligibility, access, identity, ancestry, action date and search-completeness gaps, with weighted and unweighted counts.

A report is the primary sampling unit. It may contain more than one contribution; each distinct contribution will be recorded, but a report contributes once to each report-level outcome. The primary outcome is positive when at least one frozen eligible contribution satisfies the endpoint. Contribution counts and clinically linked intervention families are separate descriptive denominators.

The target population is the eligible reports represented in the declared accessible bibliographic frame, not all worldwide orthopaedic innovation. Searches cannot capture every proprietary programme or undocumented practice. Source coverage and frame overlap will therefore be reported alongside every population interpretation.

Documentary follow-up begins at the earliest verified public date of the report version containing the contribution. The endpoint requires a later qualifying document within the horizon. Actual intervention or research dates are extracted separately. A later publication does not establish that the underlying action began after the index report.

The primary estimand includes reports with prior clinical activity, because it measures later documented use. A separate incident-use analysis includes only reports with affirmative evidence that the contribution had not yet entered patients at baseline. Unknown baseline status remains unknown; it is not assumed preclinical merely because the report contains animal or computational work.

The primary cohort ends its documentary windows at index date plus ten years, never later than 31 December 2025. Five year outcomes use index date plus five years. The recent cohort contributes only five year outcomes. Reports from 2021–2025 contribute metadata to the map, not immature ten year rates. Newly retrieved documents published in 2026 cannot establish a pre-2026 documentary event, although an explicit retrospective account may inform a separately labelled action-date history.

3 Eligibility and units of analysis

Eligible index reports present original empirical or computational work with a stated musculoskeletal application or a clearly demonstrated contribution to musculoskeletal structure, function, disease, intervention or clinical measurement. Eligible modes include in vitro experiments, animal in vivo research, human ex vivo or cadaver work, materials and device testing, physics-based modelling, machine learning, mechanistic studies and enabling methods. A preliminary human component does not itself exclude an otherwise eligible contribution.

The clinical remit includes trauma and bone healing, degenerative joint disease and arthroplasty, cartilage and orthobiologics, tendon and ligament disorders, sports injuries, spine, paediatric musculoskeletal disorders, musculoskeletal oncology and infection, foot and ankle, hand and upper limb, rehabilitation engineering, gait, prosthetics, and musculoskeletal diagnostics. General biology is eligible only when its musculoskeletal relevance is supported in the index source. Dental-only, nonmusculoskeletal and purely descriptive clinical reports are excluded.

Exclude reviews, editorials, protocols without performed original work, isolated marketing claims and duplicate versions of the same report. These remain discovery sources when useful. Include substantive conference proceedings and preprints if they contain assessable original methods and results; flag their publication status and analyse peer-reviewed reports separately. Abstract-only records may remain potentially eligible but source-limited. Non-English reports remain eligible; translation source, system and unresolved technical terminology are recorded.

Decision process

Title and abstract screening is deliberately sensitive. Both automated assessors must identify an explicit exclusion criterion to exclude a record at this stage; disagreement, unavailable abstracts or uncertain relevance proceed to source assessment. Full-source eligibility uses the same prespecified criteria. A missing full text is an unresolved eligibility decision, not an exclusion. Each excluded record receives one primary reason from an ordered list: duplicate version, outside date, outside remit, nonoriginal work, no eligible scientific contribution, or other explicitly specified reason.

Linked reports and families

Normalize DOI and PMID, preserve source identifiers, and link preprint, conference and journal versions only with verified identity. Different publications from the same experiment are linked at study level without erasing distinct report records. Every clinical benefit review uses unique studies, linking companion reports before synthesis.[8]

An intervention family requires evidence of shared technical identity and development ancestry, not an author match, common topic or similar brand. Uncertain merges remain unresolved. Report-level weighting is retained in the main cohort analysis. A family-level probability cannot be estimated from the number of observed families unless all family members in the sampling frame and their selection probabilities are established. Descriptive family maps and alternative admissible clustering sensitivities will be labelled accordingly.

4 Search architecture and coverage

The search has three distinct purposes: construct the report frame, discover downstream uses for sampled contributions, and identify clinical evidence for verified destinations. Literal searches, platform versions, execution dates, pagination, limits, counts and export hashes will be retained under PRISMA-S.[7] This protocol contains proposed strategies; no execution or completeness is claimed for them yet.

Source group	Required role and coverage condition
PubMed or Europe PMC MED	Biomedical report frame and original-source links. Europe PMC MED and PubMed are interfaces to overlapping biomedical records, not independent databases.
OpenAlex and Semantic Scholar	Complementary scholarly discovery, engineering records and citation edges. Retain each source separately before deduplication; neither is assumed exhaustive or mutually independent.
Embase, Scopus, Web of Science, Compendex, IEEE Xplore	Planned coverage extensions where authenticated access and complete exports are available. IEEE and Compendex are particularly relevant to engineering. Access is not assumed or purchased by this protocol.
ClinicalTrials.gov and WHO ICTRP sources	Registered clinical investigation, status, reported results and study identifiers; duplicate registrations linked.
Regulators and guidance	FDA device and drug records; EMA and national records where relevant; NICE, professional guidelines and arthroplasty or other clinical registries.
Patents and commercial records	Patent families, licences and company disclosures provide separate innovation signals and discovery leads. They do not establish patient use or efficacy.

Before sampling, a source-readiness report must identify which databases were actually searched and whether every page or export partition was obtained. If a planned licensed source is unavailable, disclose the omission and revise the frame claim before drawing the sample. If a required open source cannot be enumerated reproducibly, use separately defined available-source frames; do not label their union comprehensive. A sample drawn from a partial page of relevance-ranked results is unacceptable.

The broad biomedical search deliberately avoids a mandatory animal, laboratory or computational filter, and avoids NOT humans, to reduce exclusion of enabling and mixed studies. Equivalent musculoskeletal vocabulary will be translated into each supported source syntax. Query translation will be tested on development examples and documented; it will not be called independent librarian or PRESS review.

Initial frame search and the submission update use the same frozen strategy. The update should occur within 90 days of submission, with its actual date reported; this is a protocol choice rather than a claimed BMJ rule. Late indexing of an otherwise eligible old report is logged in an update frame, not silently inserted into an already sampled frame. Outcome cutoffs remain fixed unless a dated amendment changes them.

5 Sampling and allocation

The mature sample contains 12,000 records and the recent sample 4,000 records, or the full respective frame if smaller. These are record counts before eligibility assessment. No record is replaced because it is ineligible, inaccessible, uninteresting or difficult to trace. The sample size is fixed before downstream outcomes are examined.

Create strata using index publication year and four source-membership categories: biomedical only, biomedical plus engineering-oriented discovery sources, engineering-oriented discovery sources only, and uncertain membership. Membership is a retrieval attribute, not a guarantee about scientific discipline. The source categories and exact database membership rules are frozen with the frame. Uncertain identity that could create duplicate sampling requires resolution before the unified draw or separate frame-specific sampling.

Within each cohort, reserve one selected record in every nonempty stratum. Allocate the remaining slots 80% proportionally to stratum size and 20% equally across nonempty strata. Round using the largest-remainder method, with ties broken by the fixed stratum identifier. Cap allocations at stratum population size; redistribute excess by the same rule among uncapped strata until the exact total is reached. Archive the final allocation, not merely the formula.

Sampling strata use the frozen reconciled bibliographic year. The verified first-public contribution date governs eligibility and follow-up. A later date discrepancy retains its original selection probability, without replacement or silent cohort reassignment. Out-of-date records are excluded for that cohort, uncertain membership enters its bounds, and eligible spillover records are described separately. The target population is limited to the relevant bibliographic-year frame.

Generate a random seed only after the frame, allocation and protocol hashes are locked. Use a specified pseudorandom generator, archive its version and seed, and sample without replacement within each stratum. The implementation must be independently reproduced before tracing. Each report has inclusion probability n_h/N_h and expansion weight N_h/n_h. The seed will not be rerolled to improve case mix.

Previous records and blinding

Previously assessed records may enter if selected by the new random draw. Their inclusion preserves the full-frame target population. They will be flagged and reassessed under the same new protocol; prior classifications will be concealed from source packets where practicable. The new sample is not an independent replication for these records. Excluding them is a sensitivity analysis of a different population, not the primary estimate. Legacy results will never be appended to the new sample without design weights.

Frame and study dependencies

Exact report duplicates must be reconciled before sampling. Where source overlap cannot be resolved, estimates remain frame-specific with an overlap sensitivity analysis; denominators are not added. Programme and study links will support variance and influence checks. Clustering standard errors cannot transform a publication sample into a representative sample of inventions.

Operational piloting uses the existing development cohort and test fixtures. Changes to searches, extraction questions or source rules arising from development are allowed before the extension lock and are recorded. Any change after the new sample begins is a dated amendment; changes informed by outcomes are identified as such.

6 Precision and feasibility

The planned workload aims for breadth and stable descriptive estimates. It does not guarantee a particular number of eligible reports or patient-use events. At the old primary eligibility yield of 471/2,400, screening 12,000 records would yield about 2,355 eligible reports; the expanded dates and sources may have a different yield. This extrapolation is a planning scenario, not a forecast.

For an ideal simple random sample with a fully observed binary endpoint, an approximate 95% half-width d requires $n = 1.96^2\, p(1-p)/d^2$ eligible reports. Unequal weighting, source uncertainty and clustering increase the information required. The following calculations deliberately exclude those complications.

Planning prevalence	Desired half width	Eligible reports
0.5%	0.25 percentage points	3058
1%	0.5 percentage points	1522
2%	1 percentage points	753
50%	2 percentage points	2401

If 2,355 eligible reports yield an effective sample size of 1,570 after a design effect of 1.5, the approximate half-width would be 1.08 percentage points at 5% prevalence and 2.47 points at 50%. These are illustrative sampling errors only. They do not include unknown outcomes or incorrect source interpretation. A very large cohort with half its outcomes unresolved still cannot supply a precise actual-use rate.

Information priorities

The execution priority is complete application of the prespecified source procedure to the selected cohort. Additional sample size will not substitute for inadequate source scope or unresolved ancestry. No expansion will be triggered by an unexpectedly small number of successes, an unattractive plot or a nonsignificant result.

A feasibility checkpoint before the draw verifies that the source frame can be enumerated, the development workflow can retain provenance, endpoint rules can be represented without contradictions, and the required workload has an executable plan. Failure to access a source changes the declared scope; it does not silently turn unavailable evidence into a negative result.

Clinical predictor modelling is not a promised output. No model will be fitted solely because the record count is large. A separate model specification would require adequate events and outcome resolution, an explicit parameter count and model-specific sample-size calculation.[9] Under this protocol, the principal subgroup analyses are descriptive. This avoids promising a model unsupported by the three clinical links in the development cohort.

7 Contribution identity and source evidence

Before downstream searching, freeze the contribution in a concise statement: what was shown or built, the system or population, the decisive method or mechanism, the claimed function and its intended use. Record whether its role is foundational mechanism, enabling method, direct intervention or mixed. Freeze this statement from the index source before viewing downstream findings. Later corrections preserve the earlier version and reason.

Evidence class	Operational rule
A Direct contribution link	A receiving original source identifies the index contribution or its uniquely documented lineage and states or demonstrates its role in work actually performed. Identity, ancestry, action and date are all source supported.
B Supported but indirect link	Adaptation or ancestry is plausible and corroborated but one direct linkage element remains inferential. Used in sensitivity analyses and narrative mapping, not as a primary confirmed positive.
C Background or shared topic	Citation, overlapping investigators, related biology or a product family without a demonstrated role for the frozen contribution. Not sufficient for a use endpoint.
U Unresolved	Insufficient source scope, uncertain intervention version, conflicting identity, unverified action or uncertain timing. Retained explicitly.

Direct links may involve adaptation; an unchanged-copy requirement would miss legitimate translation. Record both the lineage link and every material change. A strict exact-version sensitivity accepts only unchanged material technical elements relevant to the index claim. Species or dose changes, a different algorithm version, changed geometry or an altered procedure are not silently called exact use.

The source packet contains the index source and receiving source identifiers, titles, authors, versions, retrieval dates, accessible scope, locators for the contribution and performed action, reference-list linkage when relevant, date evidence and file hashes. Original Methods, Results, a registry result or a regulator record may support different facts; an abstract supports only what it states. A secondary account can corroborate a history but cannot manufacture an unseen original Methods section.

Conflicts are resolved using the original source version relevant to the event, corrections or retractions, and explicit identity evidence. No single publisher hierarchy overrides a documented correction. A search snippet is a discovery lead unless its original passage and scope can be reliably established. Search ranking and AI confidence are never evidence grades.

Attribution is contribution use, not proof of exclusive causation. Several reports may inform one intervention. The paper will not claim that a sampled article caused a product to exist. All clinically important positive links must have a reviewable evidence packet, including limitations where a full source is unavailable.

8 Outcomes and admissible classifications

Outcome	Minimum evidence
Performed research use	A frozen contribution informs an experiment, simulation, development or validation activity that was performed. Include negative replication and critical tests; exclude proposals alone.
Patient management	The contribution informs an actual diagnostic, therapeutic or procedural decision or delivery to living patients. Clinical trial delivery qualifies, with research context recorded.
Routine care documentation	Use outside a research-only protocol, supported by a clinical registry, service record, cohort or equivalent original evidence. One commercial announcement is insufficient; extent of adoption is a separate field.
Comparative clinical evidence	A linked patient study estimates an intervention–comparator contrast, with patient-important and surrogate outcomes separated. Existence of a comparison does not establish benefit.
Patient benefit	A structured synthesis of patient-important outcomes, harms and certainty for a specific clinical question. Favourable p values, surrogate change or authorization alone cannot establish benefit.
Authorization or commercial activity	Exact product, indication, jurisdiction, pathway and date for a regulator decision; patents, licensing and commercial use recorded separately.
Informative negative finding	A performed test contradicts or constrains the indexed claim, or a source explicitly reports discontinuation, harm or failed feasibility. No search hit is not abandonment.

Prespecified binary documentary milestones at each horizon have one of three evidence states: confirmed qualifying documentation; completed bounded search without a qualifying link; or unresolved assessment. Inapplicability is a separate value used only for outcomes whose definitions genuinely do not apply, such as product authorization for a nonproduct mechanism. Inapplicability is never counted as failure. Patient-benefit and harm conclusions remain outcome-specific effects, uncertainty and certainty assessments, not a binary treatment-success endpoint. The harms-assessment milestone records systematic harms collection and reporting, separately from whether harm was detected.

Store all milestones independently. Research use is not a required precursor to observed care, and authorization is not a universal gate. Five year positives must remain positive at ten years for a cumulative documentary endpoint, while later withdrawal or refutation is stored as an additional event. An early positive is not erased by later failure.

A nonpositive classification requires all applicable search gates, completed examination of qualifying leads within the declared limits, and no unresolved lead capable of changing the endpoint. Otherwise the record remains unknown. Even a completed bounded nonpositive means only that the specified procedure found no qualifying documentation. It does not establish absence of worldwide clinical use.

9 Automated assessment and audit

All scientific screening, extraction and source interpretation will be performed by automated systems under this protocol. No human scientific reviewers are specified. The manuscript will describe separately executed AI workstreams accurately; it will not call them two independent human reviewers. Model agreement measures reproducibility, not validated accuracy. Human authors remain responsible for submitted content under BMJ policy.[10]

Two workstreams independently assess each source packet using the locked questions and receive no earlier classification. The source packet may be shared; shared evidence and related model families create correlated errors and will be disclosed. Record provider, model identifier, date, configuration, prompt hash, source packet hash and output. If a provider does not expose a model snapshot, record that limitation rather than inventing a version.

The extraction workstream records the factual fields and evidence locators. The challenge workstream tests identity, whether the action occurred, contribution ancestry, chronology, adaptation and possible contradictions. A coordinating adjudication accepts only facts supported by the packet. Unresolved disagreement produces an unknown field; majority voting alone is insufficient. All candidate patient-management positives undergo an additional challenge against the full acceptance checklist.

Random quality audit

After initial classifications are frozen, select a 20% simple random audit within cohort, initial patient-management state at the cohort's primary horizon, and mutually exclusive primary modality, using a separate archived seed. Use a mixed category when no main modality is defensible. Freeze modality rules before tracing; round audit sizes upward and retain the actual audit probabilities. Audit all serious identity conflicts and candidate clinical positives in addition. Random-audit probabilities and certainty selections are recorded separately; their yields are not pooled unweighted. Auditors reassess the same immutable source packets without the initial verdict; this audit does not receive extra discovery searches. Its outcomes are disagreement, unsupported-claim detection and correction yield, not sensitivity or specificity.

A demonstrated error is corrected. If it exposes a defective decision rule, parsing process or source interpretation pattern, rerun the corrected workflow across all eligible packets or a mechanically defined affected class independent of audit membership, and record the amendment. Do not preserve a known false classification to maintain equal effort. Additional source discovery beyond the common search procedure requires a separately logged extension applied uniformly to the affected cohort, or a separately specified probability verification design; selective deep searches may not be blended into an unqualified headline estimate.

Tests and limitations

Use deliberately altered identifiers, impossible dates, proposal-only passages, background citations and mismatched product versions to test rejection logic. Development examples test extraction consistency. These tests do not establish clinical source accuracy. Published AI benchmarks may contextualize performance but cannot validate this new ancestry task. The lack of a human or external reference standard remains a material limitation; error sensitivity analyses are mandatory. This workflow does not claim compliance with Cochrane requirements for two people extracting outcome data.[8]

10 Uniform downstream search procedure

Every eligible contribution receives the same applicable search families. Queries use the frozen contribution statement, exact identifiers, technical synonyms and documented aliases. Unsuccessful queries and no-hit results are retained with dates. Search effort is recorded as a measurement-process variable, not treated as a surrogate for truth.

Gate	Required action
G1 Identity and index source	Resolve report versions, contribution statement and earliest public date. Locate methods and results adequate for classification.
G2 Citation discovery	Enumerate available forward citation edges in both OpenAlex and Semantic Scholar. Verify cited and citing identifiers, dates and source pagination; retain failure or incomplete-export status.
G3 Contribution search	Search exact title or DOI; technical contribution plus investigators; technical contribution plus musculoskeletal indication; and its main synonyms or unique component identifiers.
G4 Clinical and critical searches	Combine contribution aliases with clinical, trial, patient, treatment, procedure, validation, replication, negative, failure, harm and discontinued terms. Search actual methods as well as titles.
G5 Applicable registers	Search trial registries for clinical candidates; regulator and patent records for identifiable products; relevant guidelines and clinical registries for care claims. Record justified inapplicability.
G6 Receiving-source assessment	Read the source passages needed to decide identity, ancestry, performed action and chronology. Backward trace every candidate qualifying lineage; inspect corrections and contradictions.
G7 Closure	Classify every retained potentially decisive lead as qualifying, nonqualifying or unresolved. Any open decisive lead blocks a searched-nonpositive classification.

For keyword discovery, inspect all retrievable results up to 200 unique records per source per contribution-query family, retaining the platform order and cutoff. For citation discovery, enumerate up to 2,000 receiving records per index report with full pagination. These are workload limits, not assumptions that later records are irrelevant. A truncated export makes each applicable endpoint that unseen records could change unresolved, even when no decisive lead appears among inspected records. Truncation cannot support a searched-nonpositive classification. Confirmed positives may still be established from sufficient evidence. Report how many records hit each cap.

For a candidate clinical destination, run the separate clinical review searches without these contribution-tracing record caps. An unavailable or capped clinical-review export yields an explicitly incomplete destination review; it cannot support a claim that no comparative evidence exists. One additional backward and forward citation generation is required for each potentially qualifying original clinical report; unresolved further decisive leads remain open.

Respect access denials and rate limits. Use lawful accessible versions and retained passages at their actual scope; do not bypass a denied route. Distinguish retryable technical failures from access restrictions. After the fixed search update, every record receives a disposition even when source access prevents resolution.

11 Appraisal of scientific foundations

The exposure of interest is the reported methodological safeguard, not an assumed latent quantity called rigor. Different designs require different validity domains. All fields distinguish reported present, explicitly absent, not reported, not applicable and not ascertainable from accessible sources. No additive cross-design quality score will be constructed.

Design	Domains and tool role
Animal experiments	SYRCLE domains for risk of bias where applicable; ARRIVE 2.0 for reporting fields. Randomization, concealment, blinding, unit of analysis, exclusions and selective reporting are separate from species and disease-model applicability.[11,12]
In vitro and materials	Independent biological versus technical replicates, comparator, batch and contamination controls where relevant, prespecified outcomes, measurement reliability, independent validation and reproducible methods. Custom domains are explicitly unvalidated.
Cadaver and biomechanics	Specimen selection, paired or repeated design, loading conditions, boundary conditions, measurement error, calibration, blinding, failure criteria, and physiological applicability.
Physics-based computation	Context of use, model influence, code and calculation verification, validation against relevant independent observations, uncertainty, sensitivity and applicability. FDA 2023 guidance informs domain selection; regulatory compliance is not inferred.[13]
Prediction and machine learning	Data provenance, patient-level separation, leakage, missingness, calibration, external validation, model version and workflow impact. PROBAST+AI supports applicable prediction-study appraisal; reporting and bias remain distinct.[14]
Linked clinical studies	RoB 2 for randomized comparisons; the final 2016 ROBINS-I for nonrandomized intervention comparisons; QUADAS-3 for diagnostic accuracy; outcome-specific certainty for coherent clinical questions.[15,16]

Freeze the exact tool documents before extraction. QUADAS-3 is the current diagnostic accuracy tool published in 2026. ROBINS-I V2 is still labelled a draft on its official site; this protocol selects the final 2016 tool rather than silently adopting a moving draft. Tool changes require an amendment and a consistent reappraisal of affected studies.

Appraisal will be automated and source-limited where appropriate. Unknown reporting cannot be treated as proof that a safeguard was absent. Historical studies will not be penalized as noncompliant with guidance that did not exist when they were performed; the extracted safeguard itself can still be described. Baseline appraisal is frozen without access to downstream outcomes where practicable. The paper will state how blinding was achieved and where it failed.

12 Clinical destination reviews

For every directly verified patient-management lineage, define the intervention version, indication, population, comparator, patient-important outcomes, harms and setting. Identify all matching clinical studies within the locked searches, including registered but unpublished studies and negative results. Searches use exact technical and commercial aliases, trial identifiers and earlier names. Related product variants are not merged automatically.

Before numerical effect extraction, freeze a dated destination-specific specification of population, exact versions, comparator hierarchy, patient-important domains, preferred instruments, follow-up windows, effect direction and multiple-report or multiple-arm rules. Disclose any prior knowledge of results; departures require a dated amendment.

The review question for each destination has two time cuts: evidence publicly available before the earliest verified patient-exposure date, where that date is identifiable, and all evidence publicly available by the relevant follow-up horizon. When exposure is only known within an interval, evidence is classified as definitely before, definitely after or temporally indeterminate. If only a publication date is known, describe evidence preceding documented use; do not call it evidence before first patient exposure.

Extract study design, setting, unique patient counts, allocation, comparator, intervention version, follow-up, outcomes, treatment effects and uncertainty, harms collection, withdrawals, sponsor and investigator overlap. Link companion papers and registrations at study level. The same trial can inform multiple sampled reports, but its participants will not be counted multiple times in clinical totals.

Benefit and harm

Patient-important outcomes include pain, function, quality of life, participation, revision or reoperation, serious adverse events and mortality where relevant. Imaging, biomarkers, tunnel dimensions, stiffness and model accuracy are labelled surrogate or technical outcomes unless their clinical role is justified. Record minimally important differences from an outcome- and population-specific validated source where available; otherwise report the effect without inventing a clinical threshold.

For each prespecified patient-important outcome, summarize effect estimates, precision, applicability and certainty. Use GRADE principles for coherent bodies of comparative clinical evidence; the automated nature and uncertainty of judgments remain explicit. Do not assign GRADE to a bibliometric transition rate. A destination may have benefit on one outcome and harm on another; no binary “successful treatment” score will collapse these distinctions.

Synthesis rules

Pool only studies addressing a clinically comparable population, intervention version, comparator, outcome and follow-up window. Prefer direct reported contrasts; use risk ratios for compatible binary outcomes and mean differences on common scales. If at least three clinically comparable studies are available, use a prespecified random-effects model with REML and Hartung–Knapp uncertainty, retaining small-study limitations. Provide a prediction interval only when its assumptions and information are adequate. Sparse or zero-event harms require an appropriate sparse-data analysis, not an automatic continuity correction.

Otherwise use structured synthesis without meta-analysis, retaining effect estimates and reasons for nonpooling.[17] Do not pool unrelated interventions into one orthopaedic treatment effect or vote-count significance. Compare animal and clinical effect sizes only for defensibly commensurable outcomes. Assess missing results against registrations and protocols; asymmetry tests are secondary and only considered with at least ten comparable studies and relevant assumptions.

13 Context funding and regulatory pathways

Baseline predictors must be measurable using information available no later than the index report. Later citations, company formation, funding and publication growth cannot be presented as baseline causes of translation. Each predictor carries its measurement date, definition, source and missingness reason.

Construct	Predeclared operational definition
Scientific category	Main clinical domain, research role and nonexclusive experimental modalities; retain mixed and unresolved classifications.
Funding	Acknowledged funding category: public, charitable, commercial, mixed, explicitly none or unknown. Extract named awards; count amounts only with an exact award linkage and relevant dates. No grant amount is assigned wholesale to a single paper.
Field activity	Number of indexed reports in the same frozen domain in the five years preceding index publication. Metadata from 2000 is needed for the earliest cohort.
Field growth	Ratio of publication counts in the latest two complete pre-index years to the preceding two, with a predefined zero-denominator rule. This measures publication activity, not market saturation.
Interest pool	Unique disambiguated active investigator groups and institutions in the preceding five years. Report entity-resolution uncertainty. Search popularity or citations are not clinical demand.
Clinical context	Available disease burden or procedure volume matched to geography and date; no imputation across unrelated populations. Absence of suitable data means the variable is omitted.
Commercial linkage	Index-date declared industry involvement, patent or licensing link with exact identity. Later commercial signals are outcomes.
Regulatory exposure	Intervention type, jurisdiction, historical rule, pathway, indication and version, each with an authoritative source.

Orthopaedics will not be assumed to be less regulated because some interventions are less invasive. Invasiveness, product risk class and applicable legal pathway are separate variables. Distinguish 510(k) clearance, De Novo authorization, PMA approval, drugs, biologics, tissue-related pathways, software and procedural practice. Current guidance cannot be applied retrospectively without verifying the relevant historical rule. FDA notes that most 510(k) clearances do not require clinical data; this does not imply absence of all oversight.[18]

IDEAL-D informs description of device evaluation and follow-up, not a mandatory sequence for every contribution.[19] An operative technique may not have a product authorization endpoint; classify it as not applicable rather than failed approval. Regulatory comparisons are restricted to comparable jurisdiction–product–period groups, and authorization never substitutes for comparative patient benefit.

Source visibility, open access, language and retrieval success will be described alongside these variables. Differences may reflect ascertainment as well as translation. No association in this observational cohort will be described as the causal effect of funding, rigor, regulation or field saturation.

14 Primary statistical analysis

Analyse the mature and recent probability samples separately. Publish the frame sizes, allocation, inclusion probabilities, dispositions and weighted eligibility totals. The confirmed-positive fraction among known eligible reports is descriptive. The primary inferential result is the lower and upper weighted report-level proportion over jointly admissible eligibility, date and outcome assignments. Unknown eligibility is retained, not discarded from the denominator. Searched-nonpositive and unknown proportions accompany the result.

Let $w_i = 1/\pi_i$, E_i denote eligibility, and $Y_i(t)$ the endpoint at horizon t. For a fully specified admissible assignment, the report-level proportion is $\Sigma w_i E_i Y_i(t) / \Sigma w_i E_i$. For uncertain records, retain a finite set of allowed joint eligibility and endpoint states. Minimize and maximize this ratio over those states, enforcing five-to-ten-year chronology and logical constraints. These are classification bounds conditional on recorded evidence and assumptions, not confidence intervals or guarantees about all actual use.

For sampling uncertainty, use conservative finite-population limits. In each stratum define four binary indicators from the admissible states: certainly eligible; possibly eligible; certainly eligible and positive; possibly eligible and positive. Invert the hypergeometric distribution for each count with a two-sided error allowance of 0.05 divided by four times the number of strata. Bonferroni coverage applies jointly across those counts. Include zero counts and singleton samples; census strata have exact observed counts.

Sum stratum count limits. A conservative outer 95% region has lower endpoint equal to the lower limit for certainly eligible positives divided by the upper limit for possibly eligible reports, and upper endpoint equal to the upper limit for possibly eligible positives divided by the lower limit for certainly eligible reports. Cap the upper endpoint at one; if its denominator limit is zero, use one. If the eligible population could be empty, report that the estimand is conditional on a nonempty eligible population. This nonsharp region is conditional on correct classification domains and does not cover unknown source error.

Report sharp joint weighted-sample classification bounds separately. Verify count inversion by exact enumeration of small populations, edge cases and independent computation. A more precise resampling method requires a separate locked specification; simulation cannot certify a nominal 95% method that materially undercovers.

Paired contrasts and shared evidence

Citation-versus-use and implementation-versus-comparative-evidence contrasts use the same reports and jointly admissible assignments. Subtracting unrelated marginal bounds is insufficient when outcomes share constraints. A clinical citation indicator requires a verified exact edge and an eligible receiving clinical publication, not an inferred reference. Report the full positive/nonpositive/unknown paired table.

The report-level comparative-evidence milestone requires a verified contribution-linked clinical destination and an eligible comparative study of a patient-important outcome, regardless of effect direction. An unverified destination does not imply no comparative evidence; unresolved identity, ancestry or review completeness produces an unknown state.

Clinical destination analyses have their own denominators: unique verified interventions, unique clinical studies or unique patients as appropriate. They are conditional on detected destinations and are not automatically representative of all interventions. For report-based analyses, describe study and programme clustering and influential programmes; use design-consistent variance methods and cluster sensitivity where identifiable. Never present clustered report estimates as invention-level probabilities.

15 Secondary analyses and interpretation

Publish prespecified descriptive strata for research role, modality, clinical domain, index year, reported safeguards, funding category and baseline field activity. Show numerator, denominator, unknown fraction and uncertainty together. Continuous baseline variables use fixed, outcome-blind quantiles from the full relevant frame where available, with boundaries archived before outcome analysis. Missingness is a visible category, not the lowest exposure level.

The proposed “Christmas tree” plots will be compact horizontal interval plots ordered from the smallest observed documented proportion at the top, with alphabetical tie-breaking and a companion fixed-order version. Ordering is descriptive and does not establish a ranking when bounds overlap. Publication year retains chronological order. Left and right panels use explicit identical denominators where intended; mirrored shapes will not imply interchangeable endpoints.

No multivariable clinical prediction model, feature selection or causal estimate is part of the core protocol. If later information justifies such work, it requires a separate dated analysis plan before model fitting, including the exact parameters, event and missingness assumptions, simulation-based adequacy, optimism control and a validation strategy. An amendment based on observed outcomes remains exploratory.

Time and sequence

Report cumulative documentation at five and ten years as the primary temporal summaries. Extract first known documentary dates and actual action intervals separately. Incident first-use analysis requires a defensible baseline without patient use and interpretable event ascertainment. Do not substitute publication time for action time, use complete cases as an unbiased survival cohort, or treat missing action dates as ordinary right censoring.

A compact timeline may display each verified clinical lineage with publication and action dates on different tracks. No survival model or competing-risk curve is promised unless the required time origin, censoring and state definitions are supported. Refutation, repurposing and discontinuation need not be terminal events. A fixed bench–trial–approval–care ladder is therefore not assumed.

Independence and transportability

Describe receiving investigator, institution and sponsor overlap using prespecified identity rules. “Independent replication” requires no overlapping named investigators or responsible development organization and a separate execution of the relevant test; uncertain organizational ties remain unknown. Independence alone is not high quality. Computational external validation, deployment, management change and patient benefit are separate milestones.

All secondary comparisons are estimation-oriented and hypothesis-generating. No p value determines whether a plot or result is reported. If a later amendment specifies formal multiple hypothesis tests, it must identify the family and multiplicity control before testing. Null findings, adverse findings and broad uncertainty receive the same reporting treatment as positive findings.

16 Missing evidence and sensitivity analyses

Uncertainty will be decomposed into eligibility, retrieval, source scope, identity, contribution ancestry, performed action, chronology, translation language and unresolved disagreement. Report how much of the cohort and weighted denominator each reason affects. Reasons may overlap. An open-access association with observed uptake may reflect documentation rather than true uptake.

Sensitivity	Prespecified implementation
Identity and adaptation	Primary direct links versus inclusion of supported indirect links; exact technical version versus documented adaptation. Unknowns stay unknown rather than becoming zero.
Eligibility and timing	Joint extreme assignments for uncertain eligibility and dates; alternative admissible first-publication dates; known versus uncertain baseline clinical activity.
Frame and source coverage	Each source frame separately; union only where deduplication is verified; peer-reviewed-only subset; language and access strata; prior-assessed overlap exclusion as a different target population.
Missing outcomes	Scenario odds ratios 0.2, 0.5, 1, 2 and 5 for positivity among unresolved versus resolved records, within sufficiently supported strata. These are assumptions, not calibrated probabilities.
Classification error	Cross a false-attribution fraction among documented positives of 0%, 1%, 5% and 10% with missed-link fractions among searched nonpositives of 0%, 1%, 5% and 10%. Retain partial identification for unknowns.
Dependence	Alternative defensible study and programme clusters; leave-one-programme-out influence; shared clinical destinations counted once in destination summaries.
Search limits	Identify capped or failed searches; show bounds with all affected records unresolved. Do not extrapolate search saturation to global recall.

The missing-outcome scenario uses the resolved-stratum proportion only as a reference. When that proportion is zero or one, or the stratum has fewer than 30 resolved reports, do not force an odds-based prediction; use an absolute unknown-positive probability grid of 0%, 1%, 5%, 20%, 50% and 100%. No imputation supplies the primary result. Inverse weighting by observed source availability cannot by itself remove nonignorable missingness.

Sensitivity surfaces will show the numerical endpoint or contrast across assumptions, including the boundary at which an interpretation changes. No region will be labelled plausible without empirical justification. Error budgets refer to expansion-weighted mass, with whole-record flips constrained by that budget; any continuous relaxation is labelled as an outer sensitivity bound. Also reverse each clinical-positive link in turn and omit each clinical destination in turn, so sparse findings are challenged even when a 10% budget cannot represent one entire case. These analyses describe vulnerability to error; they do not estimate error rates. If the qualitative conclusion changes under modest source-error assumptions, state that directly.

Absence of comparative evidence and comparative evidence of little or no benefit remain different findings. A small underpowered trial with a nonsignificant result cannot establish equivalence. An incomplete clinical review cannot justify “no evidence exists”. The clinical conclusion will refer to the evidence actually accessible and assessed.

17 Figure programme

The figure programme is designed around information that the protocol will collect. It contains five main figure candidates and supplementary diagnostics. No synthetic observations will be displayed as study results. Final selection follows the specified availability rules; visual ambition will not determine eligibility or endpoint definitions.

Figure	Data and scientific purpose	Availability rule
1 Evidence atlas	Rows show predefined role–modality groups; columns show research use, patient management, routine care documentation, comparative evidence, harms assessment and uncertainty. Linked inset shows study selection and actual frame coverage.	Always produce grouped cells with denominators; a full report-by-endpoint matrix is a searchable supplement, not thousands of illegible labels.
2 Use and evidence	Aligned horizontal estimates and bounds for documented patient management and comparative patient-important evidence, with unknown mass. Compact category panels implement the requested tree ordering.	Always show mature ten year results. Conditional clinical-destination panels use separately labelled denominators.
3 Clinical lineage timelines	Each verified lineage shows technical adaptation, documentation dates, action-date intervals, comparative evidence and harm signals. Distinguishes experimental delivery from routine care.	If ≤40 lineages, show all. If >40, show a prespecified random sample of 40 stratified proportionally by intervention type, with an archived seed, plus a complete supplementary atlas; no attractive-case selection.
4 What changes the conclusion	Sensitivity surfaces for false attribution, missed links and unresolved evidence; adjacent paired citation–implementation matrix.	Always produce the paired matrix. A contrast surface is shown only for a defined interpretable contrast, without estimated-accuracy claims.
5 Time and research value	Chronological five and ten year documentation panels plus positive, critical/negative and adapted research uses. Research outcomes may overlap and are labelled accordingly.	Use fixed horizons, not unsupported event-time curves. Show every prespecified year even when counts are small.

Visual standards

Use native vector plotting in R or Python with a shared data contract. A paper figure should fit a roughly 180 mm wide, 90–120 mm high footprint where the information permits; split panels rather than shrink text below about 8 points at publication size. Journal-specific dimensions will be checked during submission formatting. Use white backgrounds, black text, restrained teal/blue, charcoal and warm accent colours, with a distinct light neutral for unknown evidence. Do not rely on colour alone.

No cartoons, pseudoanatomy, decorative gradients, 3D plots or generated data art. No Sankey implies a mandatory sequence. Captions should state the unit, denominator, horizon, uncertainty encoding and key interpretation in roughly 45–75 words. Detailed definitions belong in the supplement. Every plotted quantity has a machine-readable source table and a reconciliation check; render at final physical size and inspect all panels.

18 Execution and completion criteria

A single orchestrated execution means a resumable workflow with immutable checkpoints, not one uninterruptible model response. Each stage has a concrete output and an acceptance test. Interrupted work resumes from validated artifacts. Search failures and source uncertainty remain visible; they do not disappear when the workflow reaches its final stage.

Stage	Required output	Completion test
0 Protocol lock	Dated protocol, searches, codebook, tool versions, planned analyses and amendment log.	Public timestamp before new draw; development exposure disclosed. No invented registration identifier.
1 Frame freeze	Raw exports, source inventory, overlap decisions, counts and hashes.	Pagination reconciles; every sampling unit has an identifier and stratum; missing sources declared.
2 Random draw	Allocation, seed, generator version, selected IDs and inclusion probabilities.	Independent reproduction yields the same sample; no replacement of difficult records.
3 Eligibility	Two automated assessments, reasons, source locators and adjudication.	Every selected record is eligible, excluded or unresolved.
4 Contributions and tracing	Frozen contributions, all applicable search gates, source packets and endpoint states.	Every eligible report has complete protocol disposition; open decisive leads remain unknown.
5 Clinical review and audit	Clinical study register, effects, appraisals, audit selection and corrections.	Every positive has a qualifying evidence packet; duplicate studies linked; errors corrected with affected-class checks.
6 Analysis and graphics	Weighted estimates, bounds, sensitivities, source tables and vector figures.	Independent arithmetic and logical checks pass; all figures reconcile to the locked data.
7 Manuscript package	Main paper, supplement, data dictionary, code, exclusions, search logs, reporting map and declarations.	Every claim traces to a result or source; documents render cleanly; author attestations supplied by actual authors.

One hundred percent completion means every selected item has received the required procedure and disposition, all planned analyses have been produced or omitted for their prespecified reason, and the manuscript package is internally consistent. It does not mean all outcomes are resolved, global use has been measured without error or a journal has agreed to publish.

A primary result may remain an interval too wide to answer a policy question. In that case the paper must describe the measurement limits and supported findings. The team may undertake a new registered extension, but cannot silently change the endpoint, remove difficult records or continue sampling until an impressive result appears.

19 Reporting governance and data release

The final manuscript should lead with the clinical question, the defined population, and the most consequential supported finding. Its main methods can be concise because full search strings, extraction rules, source scopes, appraisal decisions and analysis details are supplied in the supplement. Concision will not conceal the sampled design, unresolved evidence, automated assessments or limitations of inference.

Use PRISMA reporting for the systematic search and clinical-review components and STROBE items for the report cohort. Provide separate flow diagrams for frame construction, sampled eligibility, source tracing and clinical-study inclusion. A literature map is not a meta-analysis, and a sampled cohort is not exhaustive source assessment of every report in the map. The title and abstract will describe the actual design.

Registration and amendments

Deposit the dated protocol, search plan and analysis plan in a public versioned repository before the new draw. Seek PROSPERO registration only if the review meets its current eligibility rules and registration stage; this document does not assert acceptance or a registration number. A repository timestamp is the fallback transparency record, not retrospective preregistration of the existing study. Amendments record date, reason, affected records, whether outcomes had been seen and reruns required.

Authorship and involvement

Named human authors must supply affiliations, contributions, funding, competing interests and the guarantor statement and accept accountability. None of these can be invented. No human source-review labour is assumed in the scientific workflow. BMJ requires AI use to be disclosed and described; the methods and contributorship statements will name the actual systems and tasks.[10] Whether an editor accepts this degree of automation remains an editorial decision.

No patient or public involvement has occurred in drafting this protocol. Report that fact and the planned plain-language dissemination. Do not claim patient endorsement. Public-source meta-research normally does not involve patient recruitment, but any institutional ethics determination or exemption must be obtained and reported accurately if required; it will not be fabricated.

Data and reproducibility

Release the protocol, exact queries, source inventory, sampled identifiers, screening and exclusion decisions, contribution statements, extracted structured facts, date intervals, uncertainty reasons, permitted evidence locators, amendments, analysis and plotting code, seeds, software versions and checksums. Respect copyright: do not redistribute restricted full texts. Public exports should include identifiers and concise evidence locators sufficient for audit, with access restrictions stated.

All manuscript numbers and figures must be generated from the same locked dataset. An independent implementation will reproduce core totals and bounds; machine agreement will not be described as external validation of clinical interpretation. The final package includes both a concise paper and a technical supplement, with every omitted analysis linked to its prespecified feasibility rule.

Appendix A Proposed source search strategy

These strategies are specified for testing and translation before the frame lock; they have not been executed for this extension. The search log must retain the final exact syntax as actually accepted by each platform. Initial vocabulary is broad to avoid excluding enabling and mixed-design research.

Biomedical master query

("Musculoskeletal System"[Mesh] OR "Musculoskeletal Diseases"[Mesh] OR "Orthopedics"[Mesh] OR "Orthopedic Procedures"[Mesh] OR musculoskeletal[tiab] OR orthop*[tiab] OR bone[tiab] OR bones[tiab] OR fracture*[tiab] OR osteo*[tiab] OR cartilage[tiab] OR chondro*[tiab] OR tendon*[tiab] OR ligament*[tiab] OR menisc*[tiab] OR arthroplast*[tiab] OR intervertebral[tiab] OR "spinal fusion"[tiab] OR "spinal implant"[tiab] OR "skeletal muscle"[tiab] OR gait[tiab] OR prosthe*[tiab] OR "joint replacement"[tiab] OR "joint repair"[tiab]) AND ("2005/01/01"[Date - Publication] : "2025/12/31"[Date - Publication])

Run a corresponding 2000–2004 query for baseline field metadata only. First-publication eligibility will be adjudicated from actual source dates; the database date filter is a discovery filter. Translate MeSH and title/abstract fields explicitly for Europe PMC rather than treating PubMed syntax as interchangeable. Add documented domain-specific rescue queries for development misses before locking the strategy. No language, citation-count or positive-result filter is permitted.

OpenAlex and Semantic Scholar

Run each vocabulary family below over the same publication-year partitions and union the results. Use source-supported pagination or bulk exports and archive query parameters, returned totals, continuation tokens and export version. Search APIs differ in Boolean and wildcard support; translate each family into accepted literal queries and test it before the lock. Never copy untested PubMed operators into another API.

Family	Core discovery vocabulary
Orthopaedic remit	orthopaedic; orthopedic; musculoskeletal; biomechanics with bone, joint, spine, muscle, tendon or ligament
Bone and joint	bone; fracture; osteogenesis; osteoarthritis; cartilage; chondrocyte; arthroplasty; joint replacement
Soft tissue and spine	tendon; ligament; meniscus; intervertebral disc; spinal fusion; skeletal muscle
Engineering rescue	finite element; musculoskeletal model; implant; prosthesis; scaffold; mechanobiology; gait; wearable; robotics; additive manufacturing
Computational rescue	machine learning; deep learning; simulation; digital twin; optimization; imaging model, each combined with the musculoskeletal vocabulary

Rescue searches supplement the broad frame; their presence does not permit restricting the primary frame to fashionable methods. Licensed-source translations add controlled vocabulary where available. The final source inventory records unavailable databases and vocabulary changes.

Appendix B Search templates and retained data

Contribution tracing templates

For each frozen contribution C, index identifier I, investigator group A and documented alias V, execute and retain: exact I or title; C plus A; C plus indication; V plus clinical or patient or trial; V plus replication or validation or negative or failure; and V plus applicable registry, regulator, guideline or patent identifiers. Expand only with aliases supported by a source. Keep an alias lineage so a later product name cannot silently replace the original contribution.

Clinical review template

For each verified destination, search all technical and brand aliases with indication synonyms in the available biomedical databases and clinical registers, without a positive-result or study-design restriction. Add trial identifiers, manufacturer identifiers and exact product versions. Search harms, revision, withdrawal and discontinuation separately. Review reference lists and forward citations of eligible studies. Archive the full source-specific strings, not just this template.

Core data entities

Entity	Required fields
Report	Canonical ID, source IDs, version links, dates and intervals, frame membership, stratum, selection probability, prior-assessed flag, eligibility and reason.
Contribution	Contribution ID, report ID, frozen statement, role, modality, clinical domain, technical identity, baseline clinical status and source locators.
Source and search	Source ID, URL or DOI, version, retrieval date, scope, hash, exact query, query family, pagination, cap, access status and decisive-lead disposition.
Evidence edge	Index contribution, receiving entity, action, ancestry class, adaptation, passage locator, document date, action interval, assessor IDs and conflict status.
Endpoint	Report or contribution ID, horizon, endpoint, positive/nonpositive/unknown/not-applicable state, applicable gate completion, reason and source-edge IDs.
Clinical study	Unique study and registration IDs, intervention family, patient population, arms, outcomes, timing, effect estimates, harms, study appraisal and duplicate reports.
Audit and amendment	Initial assessment version, sampling probability, blind audit decision, discrepancy, correction reason, affected decision class, date and rerun scope.

Represent unknowns as explicit coded states with reasons, never blank numeric zeroes. Store original extraction and adjudicated values separately. Date intervals use earliest and latest supported dates and their source; incomplete year-only dates remain intervals. The schema permits several source edges for one outcome and several reports for one clinical study.

The accompanying execution package contains machine-readable planning constants, blank data tables, proposed searches, figure specifications and a validation scaffold, and tested sampling and conservative interval utilities. It does not contain a new study dataset, a completed search, a fitted analysis or production-ready database adapters.

Appendix C Reporting and review map

This map locates protocol content against PRISMA-P concepts; it is not an assertion that a journal or registry has approved the study. Author-supplied administrative information and the public registration timestamp remain outstanding until genuinely provided.

PRISMA P area	Protocol location
Title and review design	Title and Purpose; Sections 1–2
Registration and amendments	Sections 1, 5, 18–19
Authors contributions support and sponsor role	Section 19; author-supplied declarations required
Rationale and objectives	Sections 1–2
Eligibility and information sources	Sections 3–4; Appendix A
Search strategy	Section 10; Appendices A–B; exact executed searches added at lock
Data management selection and collection	Sections 7–10; Appendix B
Data items and prioritized outcomes	Sections 2, 7–8, 11–13; Appendix B
Risk of bias	Sections 9, 11–12
Synthesis and additional analyses	Sections 14–16
Meta bias and certainty	Sections 9, 12, 16
Dissemination and data availability	Sections 17–19

Independent protocol challenge

Three separately tasked AI reviews examined statistical inference, prior literature and adversarial design weaknesses during drafting. They identified the need for genuine probability sampling, disclosure of the known development results, distinct clinical benefit and uptake endpoints, design-specific appraisal, source-error sensitivity and explicit limits of AI agreement. The resulting protocol incorporates these changes. This process was methodological assistance, not external peer review or human source verification.

Decisions that cannot be delegated to invented facts

The following must be actual events or information: source access, public registration, sampling-frame size, eligible yield, observed outcomes, author identities and accountability, funding and conflicts, institutional ethics determinations, and a journal’s editorial decision. The protocol specifies how to record each; it does not assert they have occurred.

Submission positioning

The intended contribution is evidence about what research reaches patients, whether the underlying scientific contribution survives adaptation, and what clinical evidence accompanies that use. A strong general-medical paper would need adequate ascertainment and a consequential supported finding. The manuscript will present the actual result even if it supports only narrower claims. No acceptance probability or guarantee is attached to this protocol.

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Development-source provenance: Magnum_Bones_Manuscript.docx, saved version 8, and Magnum_Bones_Supplement(1).docx, saved version 6, both modified 21 October 2026 and reread for this protocol. Public methods and comparator sources were checked on 29 October 2026. This reference search was targeted protocol development, not the extension’s systematic search.